SPECIFIC REGULATIONS FOR PREPARING SENIOR DESIGN REPORTS USING LaTeX

I. REPORT STRUCTURE

The Senior Design report shall be prepared using the official LaTeX template provided by the Department. Students shall not modify the document structure, page layout, heading styles, numbering system, or other predefined formatting settings unless approved by the supervisor.

The report shall be arranged in the following order:

1. Outer cover page;
2. Inner cover page;
3. Senior Design Assignment Form;
4. Supervisor's Evaluation Form;
5. Acknowledgements (if applicable);
6. Table of Contents;
7. List of Symbols and Abbreviations (if applicable);
8. List of Figures;
9. List of Tables;
10. Abstract;
11. Main chapters of the Senior Design;
12. Conclusions and Recommendations;
13. References;
14. Appendices (if applicable).

The preliminary pages shall not be numbered as chapters. The main content shall be organized into consecutively numbered chapters, beginning with Chapter 1.

II. CONTENT OF THE Senior Design

- The report shall contain only information directly related to the assigned project topic.
- The content shall demonstrate the student's own work, understanding, analysis, design, calculation, simulation, implementation, and evaluation as appropriate to the project.
- Information obtained from books, journal papers, conference papers, standards, websites, technical documents, or other sources must be properly cited.
- Chapters, sections, and subsections shall be generated automatically using the standard LaTeX commands:

```
\chapter{...}
\section{...}
\subsection{...}
```

- Chapter, section, and subsection numbers shall not be manually entered.
- The recommended hierarchy is Chapter – Section – Subsection. Excessive subdivision of the report should be avoided.
- Unless otherwise specified by the Faculty or supervisor, the main report should normally contain between 30 and 80 pages, excluding large appendices.

III. PAGE LAYOUT AND TEXT FORMATTING

The report shall be formatted on A4 paper and printed on one side of each sheet. The page margins shall be:

Left margin:	3.5 cm
Right margin:	2.0 cm
Top margin:	2.5 cm
Bottom margin:	2.5 cm

The main text shall use a font size of 13 pt with 1.3-line spacing. A Times New Roman or equivalent Times-family typeface shall be used.

The text shall be fully justified. Students shall not use repeated spaces, manual line breaks, or repeated blank lines to position text.

Page layout and general formatting shall be controlled by the official template. Students shall not manually alter margins, font sizes, line spacing, heading spacing, or paragraph formatting for individual pages.

IV. CHAPTERS, SECTIONS, AND SUBSECTIONS

- Chapter titles shall be generated using:

  ```
  \chapter{Chapter Title}
  ```

- Section titles shall be generated using:

  ```
  \section{Section Title}
  ```

- Subsection titles shall be generated using:

  ```
  \subsection{Subsection Title}
  ```

- Numbers such as Chapter 1, 1.1, 1.1.1, etc. shall be generated automatically by LaTeX.
- Students shall not manually reproduce heading styles using commands such as `\textbf`, `\fontsize`, or `\centering` instead of the appropriate structural command.

V. PAGE NUMBERING

Cover pages and official forms at the beginning of the report shall not display page numbers.

The main body of the report shall use Arabic page numbers beginning from Chapter 1.

Page numbers shall be generated automatically by LaTeX. Students shall not manually enter page numbers.

VI. TABLE OF CONTENTS AND LISTS

The Table of Contents shall be generated automatically using:

```
\tableofcontents
```

The List of Figures shall be generated using:

```
\listoffigures
```

The List of Tables shall be generated using:

```
\listoftables
```

Students shall not manually create or edit page numbers in the Table of Contents, List of Figures, or List of Tables.

The document shall be compiled sufficiently to ensure that all page numbers, references, and lists are correctly updated.

VII. FIGURES, DIAGRAMS, AND GRAPHS

- Figures shall be clear, legible, and directly relevant to the report content.
- Figures shall be inserted using the standard `figure` environment.
- Figure captions shall be placed below the figure.
- Figures shall be numbered automatically according to the numbering system defined in the template.
- Each figure that is referenced in the text shall be assigned a unique label.

Example:

```
\begin{figure}[H]
    \centering
    \includegraphics[width=0.8\textwidth]
                    {figures/system_model.pdf}
    \caption{System model}
    \label{fig:system-model}
\end{figure}
```

Figures shall be referenced in the text using:

```
Figure~\ref{fig:system-model}
```

Students shall not manually type figure numbers.

Figures reproduced or adapted from other sources shall include an appropriate citation.

Whenever possible, vector graphics or high-resolution images should be used. Low-quality screenshots should be avoided.

VIII. TABLES

Tables shall be prepared using appropriate LaTeX environments such as `table`, `tabular`, `tabularx`, or `longtable`.

Table captions shall be placed above the table.

Example:

```
\begin{table}[H]
    \centering
    \caption{System parameters}
    \label{tab:system-parameters}

    \begin{tabular}{ccc}
        \hline
        Parameter & Symbol & Value \\
        \hline
        Resistance & $R$ & 0.7 $\Omega$ \\
        Inductance & $L$ & 2.72 mH \\
        \hline
    \end{tabular}
\end{table}
```

Tables shall be referenced using:

```
Table~\ref{tab:system-parameters}
```

Students shall not manually enter table numbers.

Tables should be created directly in LaTeX rather than inserted as screenshots whenever practicable.

IX. EQUATIONS AND MATHEMATICAL EXPRESSIONS

Mathematical expressions shall be typeset directly using LaTeX. Images or screenshots of equations shall not be used.

Short mathematical expressions appearing within a sentence shall use inline mathematical mode.

Example:

```
$R = 0.7\,\Omega$
```

Important equations shall be placed in a numbered equation environment.

Example:

```
\begin{equation}
    J\frac{d\omega}{dt}
    = T_e - T_L - B\omega
    \label{eq:mechanical-model}
\end{equation}
```

Equations shall be referenced using:

```
Equation~\eqref{eq:mechanical-model}
```

Equation numbers shall not be manually entered.

Mathematical notation shall be consistent throughout the entire report.

X. SYMBOLS, UNITS, AND ABBREVIATIONS

- SI units shall be used wherever applicable.

- Symbols and units shall be written consistently throughout the report.
- The `siunitx` package should be used for physical quantities and units where appropriate.

Examples:

```
\SI{220}{\volt}
\SI{2.72}{\milli\henry}
\SI{450}{\radian\per\second}
```

Symbols and abbreviations frequently used in the report shall be included in the List of Symbols and Abbreviations.

Each abbreviation should normally be defined when it first appears in the text.

XI. SOURCE CODE

Source code that is necessary for explaining an algorithm, controller, simulation, or implementation may be included in the main report.

The `listings` environment, or another environment defined by the official template, shall be used.

Example:

```
\begin{lstlisting}[language=Matlab]
A = [-R/L 0;
      0  -R/L];

K = hinfsyn(P,nmeas,ncont);
\end{lstlisting}
```

Long source-code listings that are not essential to the immediate discussion should be placed in the appendices.

Source code should not normally be inserted as screenshots.

XII. CROSS-REFERENCING

Chapters, sections, figures, tables, equations, and other numbered elements shall be cross-referenced using the standard LaTeX mechanism.

The following commands shall be used where appropriate:

```
\label{...}
\ref{...}
\eqref{...}
```

For example:

```
As shown in Figure~\ref{fig:response}, ...
```

Students shall not manually type a reference number such as “Figure 3.2” when the number can be generated automatically.

Cross-referencing ensures that all numbers remain correct when figures, tables, equations, or sections are added or removed.

XIII. REFERENCES AND CITATIONS

All sources used in preparing the Senior Design shall be properly cited.

References shall be managed using the bibliography system provided with the official template. Bibliographic information shall normally be stored in the file:

```
References.bib
```

A reference may be cited in the text using:

```
\cite{reference-key}
```

Students shall not manually enter reference numbers such as “[1]”, “[2]”, etc.

Bibliographic entries shall contain sufficient information to identify the original source.

Titles of books, journal articles, conference papers, standards, and other publications in foreign languages shall be retained in their original language and shall not be translated solely for the reference list.

All ideas, data, figures, tables, equations, algorithms, or results obtained from another source shall be appropriately acknowledged.

XIV. APPENDICES

Supporting material that is useful but too extensive for the main body of the report should be placed in the appendices.

Appendices may include:

- complete source code;
- detailed calculation sheets;
- large data tables;
- technical drawings;
- additional simulation results;
- equipment specifications;
- supplementary design documents.

Appendices shall be created using the appendix environment defined in the official template.

Example:

```
\begin{appendices}

\chapter{MATLAB SOURCE CODE}

...

\chapter{SYSTEM PARAMETERS}

...

\end{appendices}
```

Appendix titles and numbering shall not be manually entered.

XV. LaTeX PROJECT ORGANIZATION

Students shall use the official LaTeX template provided by the Faculty.

The main formatting commands contained in the document preamble shall not be modified unless approval is obtained from the supervisor.

A recommended project structure is:

```
SeniorDesign/
|
|-- main.tex
|-- References.bib
|
|-- figures/
|   |-- figure01.pdf
|   |-- figure02.png
|   `-- ...
|
|-- chapters/
|   |-- chapter1.tex
|   |-- chapter2.tex
|   `-- ...
|
`-- appendix/
    `-- ...
```

File and folder names should preferably contain Latin letters, numbers, hyphens, or underscores.

Spaces, Vietnamese diacritics, and unnecessary special characters in file names should be avoided where possible.

Recommended examples include:

```
motor_model.pdf
chapter_01.tex
simulation_result.png
```

XVI. DOCUMENT COMPILATION AND FINAL CHECK

Before submission, the complete LaTeX project shall compile successfully and produce the final PDF report.

The final document shall not contain unresolved references such as:

```
??
[?]
```

Before generating the final PDF, students shall check:

- the Table of Contents;
- page numbering;
- chapter and section numbering;

- figure numbering;
- table numbering;
- equation numbering;
- cross-references;
- citations;
- bibliography;
- appendices;
- font display;
- mathematical symbols;
- image quality.

The document may be compiled using TeX Live, MiKTeX, or another compatible LaTeX environment. Regardless of the compilation environment, the resulting PDF shall conform to the official Department template. Students are not allowed to use Overleaf or other online LaTeX editors to protect the knowledge and intellectual property of the project.

XVII. SUBMISSION REQUIREMENTS

The printed report shall satisfy the following requirements:

- A4 paper;
- single-sided printing;
- one bound copy, unless otherwise specified;
- soft cover;
- formatting in accordance with the official template.

The electronic submission should include:

- the final PDF report;
- the complete LaTeX source files;
- the `References.bib` file;
- all figures and graphics;
- all additional files required to compile the document.

The submitted source project shall be complete and capable of being recompiled to produce a PDF corresponding to the submitted report.

XVIII. ELIGIBILITY FOR SENIOR DESIGN DEFENSE

A student shall be eligible to defend the Senior Design when the following requirements have been satisfied:

1. The Senior Design has been completed within the specified period.
2. The report satisfies the academic and technical requirements of the assigned project.

3. The report complies with the official LaTeX template issued by the Faculty.
4. The complete LaTeX source compiles successfully without errors affecting the document content or presentation.
5. The final PDF and required source files have been completed and submitted.
6. All external sources have been appropriately cited and the reference list has been properly generated.
7. The supervisor has approved the student to proceed to the Senior Design defense.

Note: Students are responsible for the academic and technical content of the report. The official LaTeX template is responsible for the standardized document formatting. Students should therefore focus on entering and organizing the content correctly rather than manually modifying the predefined formatting parameters.
Note: This section is provided for guidance only and is included in the main document using the command `\input{Regulations.tex}`. Students shall remove or comment out this command before submitting the final report and shall prepare the report content in accordance with the requirements specified in this template.

THAI NGUYEN UNIVERSITY OF TECHNOLOGY
FACULTY OF INTERNATIONAL TRAINING
ENGINEERING AND TECHNOLOGY DEPARTMENT

ELECTRICAL ENGINEERING

SENIOR DESIGN

Project Title

Author: Student's name 1 - Student's ID
Student's name 2 - Student's ID

Supervisor: Supervisor's name

Thai Nguyen - 2026

THAI NGUYEN UNIVERSITY OF TECHNOLOGY
FACULTY OF INTERNATIONAL TRAINING
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THAI NGUYEN UNIVERSITY
OF TECHNOLOGY

Faculty of International Training

SOCIALIST REPUBLIC OF VIETNAM

Independence – Freedom – Happiness

SENIOR DESIGN

COURSE: .....

DEPARTMENT: Engineering and Technology

Student: Student's name 1 - Student's ID
Student's name 2 - Student's ID

Class: Class name

Major: Major name

Assignment date: dd/mm/yyyy

Completion date: dd/mm/yyyy

1. Project title

.....

.....

VICE HEAD OF DEPARTMENT

(Signature and full name)

SUPERVISOR

(Signature and full name)

Composition of the committee:

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LIST OF SYMBOLS AND ABBREVIATIONS

(Example of the table of symbols and abbreviations)

No	Symbol / Abbreviation	Unit	Definition
1	AC	-	Alternating Current
2	DC	-	Direct Current
3	V_{dc}	V	Direct Current Voltage
4	R	Ω	Resistance
5	L	H	Inductance
6	ω	rad/s	Angular velocity
7	T_e	N.m	Electromagnetic torque
8	J	kg.m^2	Moment of inertia
9	i	A	Current

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ABSTRACT

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